

DS & AI

Artificial Intelligence

Informed Search

DDP 01 Discussion Notes

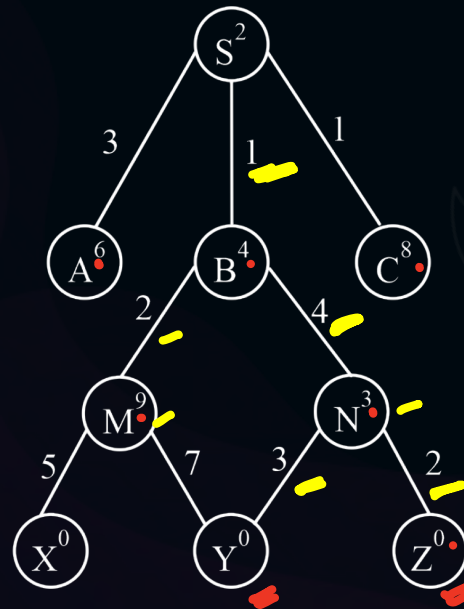


By- Siddharth sir

#Q. Consider the search space depicted in the Figure below. S is the initial state. X, Y, an Z are the states that satisfy the goal test.

Use alphabetical order of nodes to break ties. Let P denote the set of states present in the optimal path using A* algorithm and Q represent the set of states present in the optimal path using GBFS. The cardinality of $P \cap Q$ is

3



(S → B → N → Y)

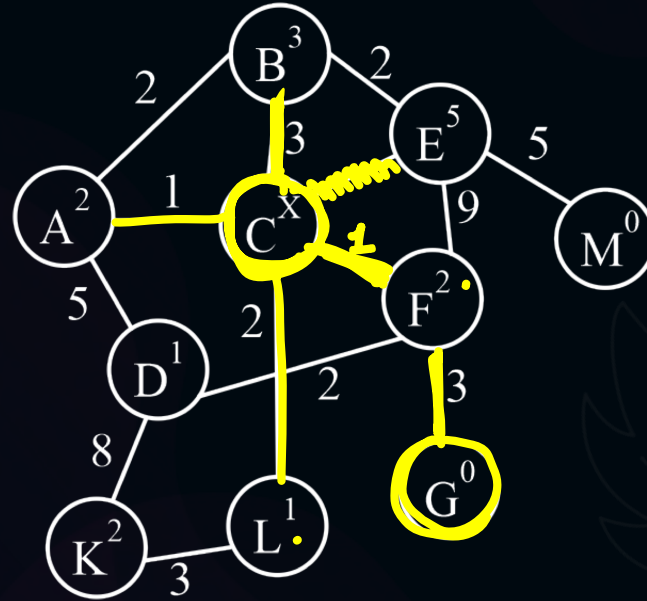
	S	B	N	Z
S	0	0	0	2
A	9	9	9	
B	5	5	5	
C	9	9	9	
M	∞	12	12	
N	∞	8	8	
X	∞	8	8	
Y	∞	8	8	
Z	∞	∞	7	

$$(S \rightarrow B \rightarrow N \rightarrow Z)$$

1 So ③



#Q. What can be the ~~minimum~~ ^{maximum} value of 'x' possible to keep the graph consistent?



* Consistent

$$h(N) \leq C(N, P) + h(P)$$

$$C \rightarrow F, L, A, B$$

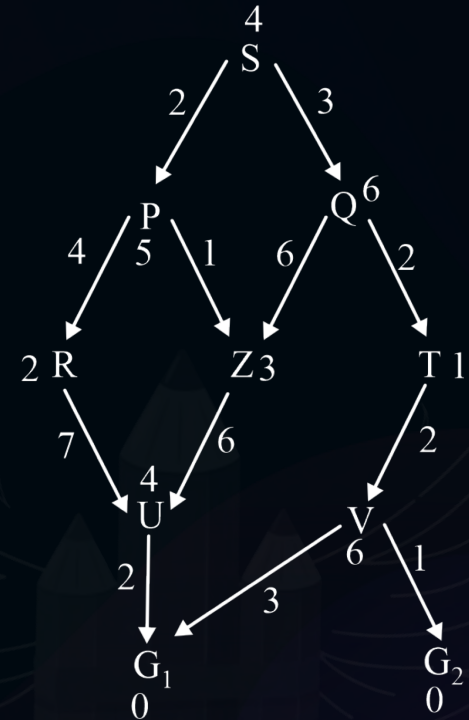
$$h(C) \leq C(C \text{ to neighbour}) + h(\text{neigh})$$

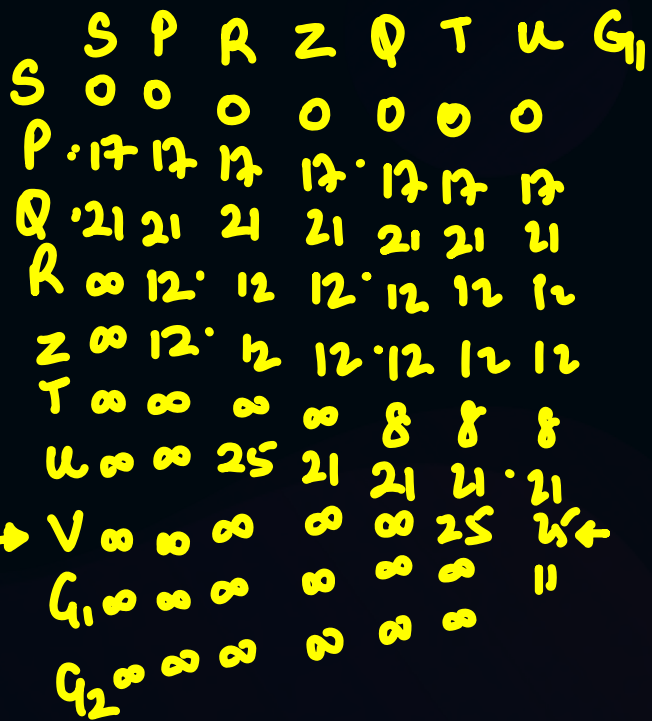
$$\left. \begin{array}{l} x \leq 6 \\ x \leq 3 \\ x \leq 3 \\ x \leq 3 \end{array} \right\} \Rightarrow \textcircled{3}$$

#Q. You are performing Weighted A* search on the following graph where the start node is S, and the goal nodes are G1 and G2.

The f-value is calculated using the formula:
 $f(n) = g(n) + w \times h(n)$ where the weight w for the heuristic is set to 3.

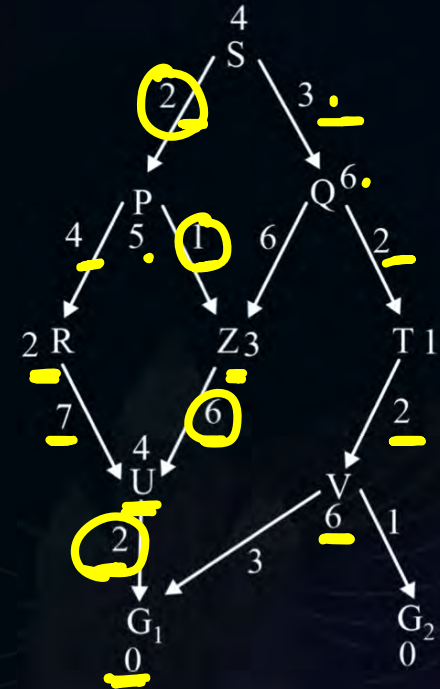
Calculate the size of the **Close list** after the completion of the entire process.





SPRZQTUG,

⑧ Close list
① open list



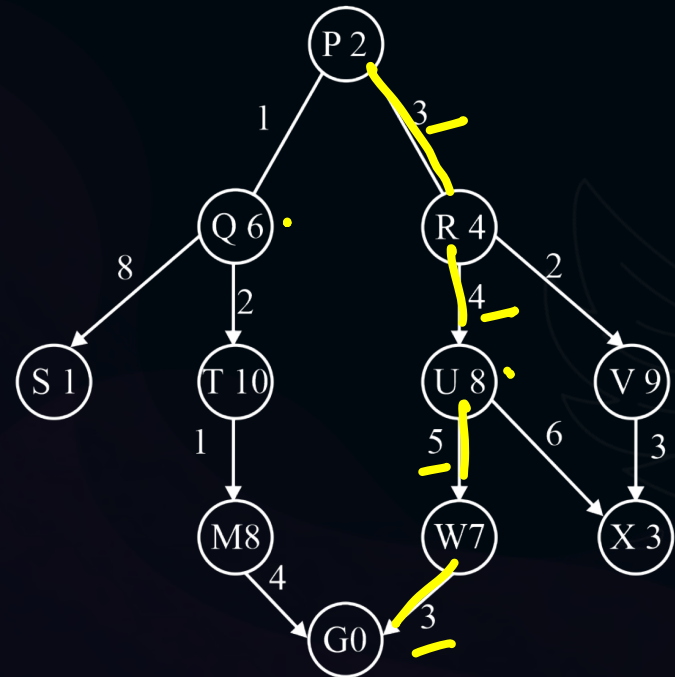
#Q. The branch factor for a graph is 5 and the depth threshold is 3. The difference between the worst case time and space complexity if Iterative Deepening A* search is used will be .

$$T_c \Rightarrow O(b^d) \Rightarrow 5^3 = 125$$

$$S_c \Rightarrow O(bd) \Rightarrow 15$$

$$\text{diff} = 125 - 15 = \underline{\underline{110}}$$

#Q. What will be the path cost in reaching the goal node G starting from node P, when Breadth First Heuristic Search algorithm is applied on the following graph? Note: Use alphabetical order of nodes to break ties.



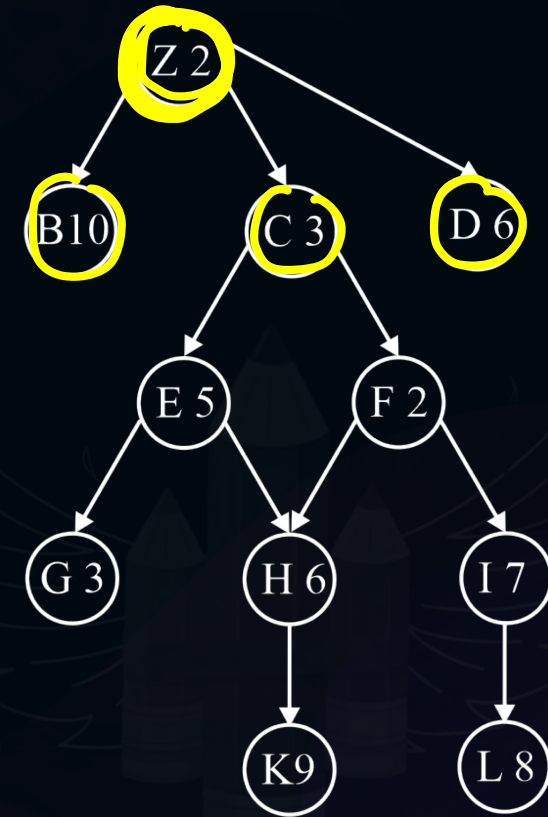
BFHS PRQSUXWG

~~RRQS~~ ~~UX~~ ~~XT~~ ~~VT~~

PRUWG

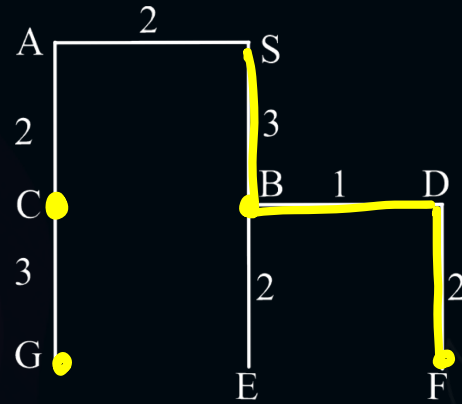
15

#Q. In a Hill Climbing problem, we have a graph where the nodes are represented by letters, and each letter is associated with a number based on its position in the alphabet (A = 1, B = 2, C = 3, ..., Z = 26). Z is the start node and G, K, and L are the goal nodes. Determine the code (number) corresponding to the letter on which the Hill Climbing algorithm will stop.



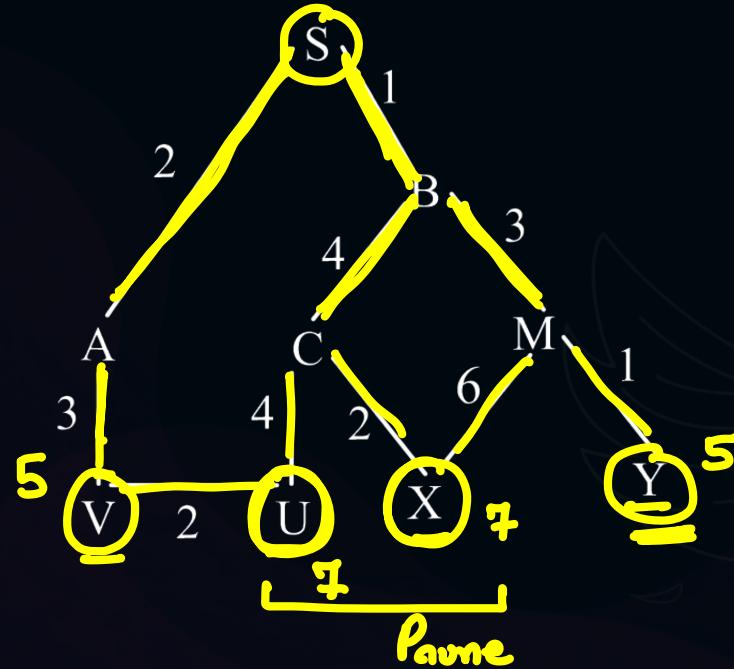
$Z \rightarrow \text{stop @ } Z \rightarrow (26) \checkmark$

#Q. Calculate the bound of the following graph where S is the Start node and G and F are goal nodes.



$S \rightarrow B \rightarrow D \rightarrow F \Rightarrow 6 \checkmark$

#Q. Count the number of nodes that will be pruned when branch and bound algorithm is applied on the following graph. S is the start node, U, V, X and Y are end nodes.





	P	Q	R
P	0	0	0
Q	3	3	3
R	5	5	5
S	∞	∞	7x
U	∞	9x	
T	∞	∞	8x

Next $t_k = \min(\text{P value})$

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#Q. Let b denote the branching factor and d denote the maximum depth of a graph.

The difference between the worst case time complexity and worst case space complexity on this graph using BFHS is ____.

$$T_c = S_c = O(b^d)$$



THANK - YOU

